together than in the standard cosmological model, making it possible for a common temperature to be easily established. Then, through Guth's momentary burst of cosmological inflation—followed by the more usual expansion of the standard cosmological model—these regions of space were able to become separated by the vast distances we witness currently. And so, the brief but profound inflationary modification of the standard cosmological model solves the horizon problem (as well as a number of other important problems we have not discussed) and has gained wide acceptance among cosmologists.3

We summarize the history of the universe from just after the Planck time to the present, according to the current theory, in Figure 14.1.

Cosmology and Superstring Theory

There remains a sliver of Figure 14.1, between the big bang and the Planck time, that we have not yet discussed. By blindly applying the equations of general relativity to that region, physicists have found that the universe continues to get ever smaller, ever hotter, and ever denser, as we move backward in time toward the bang. At time zero, as the size of the universe vanishes, the temperature and density soar to infinity, giving us the most extreme signal that this theoretical model of the universe, firmly rooted in the classical gravitational framework of general relativity, has completely broken down.

Nature is telling us emphatically that under such conditions we must merge general relativity and quantum mechanics—in other words, we must make use of string theory. Currently, research on the implications of string theory for cosmology is at an early stage of development. Perturbative methods can, at best, give skeletal insights, since the extremes of energy, temperature, and density require precision analysis. Although the second superstring revolution has provided some nonperturbative techniques, it will be some time before they are honed for the kinds of calculations required in a cosmological setting. Nevertheless, as we now discuss, during the last decade or so, physicists have taken the first steps toward understanding string cosmology. Here is what they have found.

It appears that there are three essential ways in which string theory modifies the standard cosmological model. First, in a manner that current research continues to clarify, string theory implies that the